

My IFSCA Grade A (IT) Preparation Journey

Exam Structure and Strategy Overview

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IFSCA Grade A (IT Stream) – Exam Structure

Selection Process:

Phase I → Phase II → Coding Test → Interview

Phase I – Online Screening Examination

Paper	Marks	Duration
Paper 1 (Reasoning, Quantitative Aptitude, English, General Awareness)	100	60 min
Paper 2 (IT Specific – Objective)	100	60 min

Negative marking: 1/4th per incorrect answer. Marks are used only for shortlisting to Phase II.

Phase II – Online Main Examination

Paper	Marks	Type
Paper 1 (Essay, Precis, Comprehension)	100	Descriptive
Paper 2 (Advanced IT – Objective)	100	Objective

Weightage in Final Merit:

- Paper 1 – 1/3
- Paper 2 – 2/3

Negative marking applicable in Paper 2 only.

Coding Test (IT Stream Only)

The Coding Test is **qualifying in nature**. Failure to qualify results in immediate elimination from the selection process, irrespective of Phase II performance.

In our cycle, out of 19 candidates shortlisted for interview, only 12 were finally interviewed. The remaining candidates were disqualified at the Coding Test stage.

Interview

Final Merit Composition:

- Phase II – 85%
- Interview – 15%

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1 Phase I Preparation Strategy

1.1 Paper 1

1.1.1 Quantitative Aptitude

For Quantitative Aptitude, I took guidance from Kaushik Mohanty through his YouTube channel.

[Career Definer](#)

His lectures focus on conceptual clarity along with practical shortcut techniques that are extremely useful for time-bound examinations.

Aspirants can follow his channel to strengthen fundamentals and learn efficient tricks required to solve questions quickly and accurately under exam pressure.

1.1.2 Reasoning

For Reasoning preparation, one may follow Saurav Sir (Faculty at Adda247). The following playlist was helpful for covering standard patterns and improving approach towards puzzles and logical reasoning:

[Reasoning Playlist by Saurav Sir \(Adda247\)](#)

While I did not score exceptionally high in the Reasoning section, I primarily relied on the above resource for understanding question patterns and practicing regularly. Aspirants are advised to focus heavily on puzzle-based questions and maintain consistent timed practice.

1.1.3 English

For English preparation, one may follow Nimisha Bansal's YouTube channel:

[Nimisha Bansal – English Preparation](#)

Her lectures are particularly useful for improving grammar fundamentals, comprehension skills, and structured preparation for descriptive components such as essay and precis writing. Regular reading practice along with guided learning from her sessions can significantly enhance clarity and writing quality.

1.1.4 General Awareness

I did not attempt any questions from the General Awareness section in Phase I. Hence, I do not consider myself the right person to suggest preparation resources for this section.

Note: I strategically backed my strengths and focused primarily on Quantitative Aptitude and English. This approach was sufficient to secure 30+ marks in Paper 1. My broader strategy was to compensate by targeting higher marks in Paper 2 (IT Stream), where I aimed to gain the additional margin required for comfortable qualification.

1.2 Paper 2 (IT Stream)

Structured Practice Framework

Across all technical subjects, I followed a uniform reinforcement strategy to ensure depth and exam-readiness.

For each micro-topic, I generated a large set of exam-oriented MCQs aligned with the SEBI IT pattern: The prompt framework I consistently used was:

```
"I am preparing for IFSCA Grade A (IT) exam whose syllabus aligns with SEBI IT.
Generate 100 MCQs with 4 options and provide answers with detailed explanations,
following the SEBI IT pattern."
```

This structured, repetition-driven reinforcement ensured comprehensive conceptual clarity across subjects.

1.2.1 Database Management Systems (DBMS)

For DBMS preparation, I followed a dedicated playlist by Vishwadeep Gothi:

[DBMS Playlist – Vishwadeep Gothi](#)

The playlist was sufficient to build strong foundational clarity.

The structured practice framework described earlier was applied to every DBMS micro-topic, including ER Model, Relational Model (Relational Algebra, Tuple Calculus) etc. :

1.2.2 SQL Queries

For SQL Queries, I relied primarily on the playlist by Vishwadeep Gothi:

[DBMS Playlist – Vishwadeep Gothi](#)

This helped develop clarity in query construction and logical flow.

The same structured practice framework was applied here, with particular emphasis on query-based PYQs asked in SEBI IT examinations.

1.2.3 Programming Concepts (Java / C / C++)

I did not undertake additional dedicated preparation for Programming Concepts, as I was already comfortable with Python, C, and C++ from my academic background.

However, the examination emphasizes conceptual and theoretical clarity rather than pure coding ability.

Special focus areas included:

- Variable scope (local, global, static)
- Lifetime of variables
- Memory allocation (stack vs heap)
- Pointers and references
- Function calls and recursion
- Object-oriented concepts (Java/C++)

Even in familiar subjects, I reinforced theoretical understanding using the same structured practice methodology described earlier.

1.2.4 Data Analytics / Python

For Python-related concepts and data analytics foundations, I followed the following.

[Python Data Analytics Playlist](#)

My focus was on the first 12 videos covering topics up to File Handling, including:

- Python basics and data types
- Control structures
- Functions and modular programming
- File handling

Each micro-topic was strengthened using the structured practice framework to ensure exam-oriented conceptual clarity.

1.2.5 Algorithms for Problem Solving

This section is highly conceptual and tests clarity of algorithm selection, data structure usage, and time/space complexity analysis rather than heavy implementation.

Tree and Graph Traversals

- Depth First Search (DFS)
- Breadth First Search (BFS)

Preparation should focus not only on implementation but also on:

- Structural differences between DFS and BFS
- Underlying data structures (Stack vs Queue)
- Time and Space Complexity
- When to prefer one over the other

A comparison-based understanding is crucial, as exam questions often test algorithm selection rather than code writing.

Connected Components

- Undirected graphs
- Directed graphs

Key topics include:

- Strongly Connected Components (SCC)
- Weakly Connected Components (WCC)
- Kosaraju's Algorithm
- Tarjan's Algorithm

Focus areas:

- Difference between SCC and WCC

- Algorithmic approach for each
- Time complexity comparison (Kosaraju vs Tarjan)
- Conceptual flow of graph transposition (in Kosaraju)

Comparative clarity between these algorithms is more important than memorization.

Spanning Trees

- Spanning Tree
- Minimum Spanning Tree (MST)
- Kruskal's Algorithm
- Prim's Algorithm

Preparation emphasis:

- Greedy strategy differences
- Edge-based vs Vertex-based expansion
- Role of Disjoint Set Union (Union-Find)
- Time complexity analysis

Hashing For hashing, I followed Vishwadeep Gothi's lectures (Videos 40, 41, 42):

[Hashing Lecture – Vishwadeep Gothi](#)

Focus areas:

- Hash functions
- Collision handling (Chaining, Open Addressing)
- Load factor
- Time complexity in average vs worst case

Shortest Path Algorithms

- Dijkstra's Algorithm
- Bellman-Ford Algorithm
- Floyd-Warshall Algorithm
- Johnson's Algorithm

Key understanding required:

- Handling of negative weights
- Single-source vs All-pairs shortest path
- Time complexity comparison
- When each algorithm is preferred

Greedy Algorithms

- Activity Selection
- Fractional Knapsack
- Huffman Coding
- Prim's MST
- Kruskal's MST
- Dijkstra's Algorithm
- Job Scheduling

The focus should be on recognizing the greedy choice property and optimal substructure.

Dynamic Programming (1D Array Based)

- State definition
- Transition relation
- Iterative vs Recursive DP
- Time and space optimization

Exam-Oriented Focus In most cases, the examination tests:

- Which data structure is used by a given algorithm
- Time Complexity (TC)
- Space Complexity (SC)
- Algorithm selection under constraints

Note: For conceptual clarity, I referred to *Striver's algorithm lectures* by searching the specific algorithm name and preparing structured notes.

Thereafter, I applied the structured practice framework described earlier to reinforce each algorithm through objective-question based revision.

1.2.6 Computer Networks

For Computer Networks, I followed the Gate Smashers playlist, which was sufficient to develop strong conceptual clarity and confidently attempt exam-level questions.

[Computer Networks – Gate Smashers Playlist](#)

The preparation strategy focused on both conceptual depth and protocol-level details, as the examination often tests precise technical distinctions.

High-Priority Areas

- OSI and TCP/IP Models
- Transport Layer protocols (TCP vs UDP)
- Application layer protocols (HTTP, FTP, SMTP, DNS)
- Standard Port Numbers
- Flow control and congestion control
- Error detection and correction

Protocol-Level Depth Particular attention should be given to:

- Active vs Passive mode in FTP
- Port numbers used by different protocols
- Connection establishment and termination in TCP
- Differences between connection-oriented and connectionless communication

The examination may test fine-grained distinctions, such as which port number is used for a specific service or how control and data channels differ in FTP.

Firewall (Extremely Important) Firewall is a high-weightage topic and should be prepared thoroughly. Focus on:

- Packet filtering firewall
- Stateful inspection firewall
- Proxy firewall
- Application layer firewall
- Differences between IDS and IPS

Conceptual clarity regarding how firewalls operate at different OSI layers is essential, as questions are often framed around real-world security scenarios.

Exam-Oriented Approach Computer Networks questions are typically conceptual and detail-oriented. Emphasis should be placed on:

- Precise definitions
- Protocol comparisons
- Port numbers
- Layer-specific functionalities
- Practical distinctions (e.g., Active vs Passive FTP)

The structured practice framework described earlier was applied to reinforce protocol-level clarity and objective-question accuracy.

1.2.7 Information & Cyber Security Concepts

For Information and Cyber Security, my preparation focused on developing strong conceptual clarity regarding major attack vectors, defense mechanisms, and security architecture fundamentals.

The emphasis was on understanding:

- Types of cyber attacks
- Vulnerabilities and threat models
- Cryptographic foundations
- Network security mechanisms
- Secure authentication and access control

Major Attack Categories Covered

- Phishing and Social Engineering
- SQL Injection
- Cross-Site Scripting (XSS)
- Man-in-the-Middle (MITM)
- Denial of Service (DoS / DDoS)
- Brute Force and Dictionary Attacks
- Replay Attacks
- Session Hijacking

Preparation involved understanding:

- How each attack works
- Which layer it targets
- Real-world scenarios
- Prevention and mitigation techniques

Core Security Concepts Special attention was given to:

- CIA Triad (Confidentiality, Integrity, Availability)
- Encryption vs Hashing
- Symmetric vs Asymmetric Cryptography
- Digital Signatures
- Public Key Infrastructure (PKI)
- Authentication vs Authorization
- Multi-Factor Authentication (MFA)

Exam-Oriented Focus Questions in this section are typically conceptual and scenario-based. The examination often tests:

- Identification of the type of attack
- Appropriate mitigation technique
- Security mechanism used in a given scenario
- Difference between similar-looking concepts

Accordingly, the preparation emphasized clarity of attack mechanics and defense strategies, along with objective-question based reinforcement.

You can use Chat Gpt for making notes on it and can take help from random yt videos.

1.2.8 Data Warehousing

For Data Warehousing, I referred to a focused lecture by Vishwadeep Gothi, which provided clarity on core concepts and analytical operations.

[Data Warehousing – Vishwadeep Gothi Lecture](#)

Core Concepts and Schema Types

- Data Warehouse Architecture
- Star Schema
- Snowflake Schema
- Fact and Dimension Tables
- Data Cubes and Multidimensional Structures

Multidimensional Operations Understanding the key OLAP operations is crucial, including:

- **Slice** — Selecting a single layer from a cube
- **Dice** — Selecting a sub-cube by specifying multiple dimensions
- **Drill-Down** — Navigating to more detailed data
- **Roll-Up** — Aggregating data to a higher level
- **Pivot** — Reorienting the multidimensional view
- **Aggregation** — Summarizing measure values

Data Mart and Its Techniques

- Data Mart concepts and types
- Difference between enterprise data warehouse and data mart
- Design considerations for departmental analytics

Design Techniques Familiarity with the following modeling approaches is important:

- Top-Down vs Bottom-Up approaches
- Dimensional Modeling
- ER-to-Dimensional transformations
- Schema normalization considerations

Exam-Oriented Focus Questions in this section typically test:

- Differences between schema types (Star vs Snowflake)
- The purpose and application of OLAP operations
- Suitable modeling techniques for specific analytical requirements
- Distinctions between data warehouse and data mart

Reinforcement through objective practice was applied using the structured framework described earlier.

1.2.9 Shell Scripting & UNIX Programming

For Shell Scripting, I referred to the following lecture by Prashant. Watching the first 2–3 hours from the start provides sufficient coverage of core concepts and practical examples:

[Shell Scripting – Prashant Lecture](#)

Key focus areas included:

- Shebang and script execution
- Variables and parameter expansion
- Control structures (if, loops, case)
- Functions and script modularity
- I/O redirection and pipes

For UNIX Programming fundamentals, I followed another lecture by the same instructor:

[UNIX Programming – Prashant Lecture](#)

Important topics covered were:

- File and directory operations
- Permissions and process control
- Shell utilities (grep, awk, sed)
- Filters and text processing

These lectures provided conceptual clarity and practical exposure suitable for objective-style questions.

1.2.10 Artificial Intelligence / Machine Learning

For AI/ML preparation, one may follow *CampusX* lectures for strong conceptual understanding. The coverage is comprehensive and builds clarity from fundamentals to applied concepts.

If time is limited, *Krish Naik's* lectures can serve as a quicker alternative for revision-oriented preparation.

Since the examination primarily asks theoretical and conceptual questions, in-depth mathematical derivations are not necessary. Focus areas should include:

- Supervised vs Unsupervised learning
- Bias–Variance tradeoff
- Overfitting and Regularization
- Evaluation metrics (Precision, Recall, F1, ROC)
- Basic understanding of popular algorithms (Regression, SVM, KNN, Decision Trees)

Occasionally, framework-specific theoretical questions (e.g., TensorFlow or PyTorch concepts) may be asked. For such cases, official documentation can be referred to. However, the return on time investment for deep framework study may not be optimal unless ample preparation time is available.

Overall, strong conceptual clarity is sufficient for this section, and structured revision is more important than implementation-level depth.

2 Coding Questions Strategy

The coding syllabus was largely aligned with SEBI IT Phase 2 (Paper 2). The questions were implementation-based and focused on core data structures and fundamental algorithms rather than advanced competitive programming concepts.

The following types of questions were asked:

1. Write a program for Bubble Sort from scratch (without using inbuilt sort functions).
2. Merge two sorted arrays and return the final sorted array (consider only valid non-zero elements).
3. Construct a Binary Search Tree (BST) from a given array and print its inorder traversal.
4. Construct a BST from a given array and compute its height.
5. Check whether a given string s occurs in another string t (substring check).
6. Implement Fibonacci using Dynamic Programming.
7. Find the minimum number of coins required (DP approach, not greedy).

Preparation Strategy

If you solve the Easy and Medium problems from Striver's DSA Sheet, it will be more than sufficient for this section.

Priority Areas

- Binary Trees (BT)
- Binary Search Trees (BST)
- Basic Sorting
- String Manipulation
- 1D Dynamic Programming

Trees (both BT and BST) should receive special attention, as multiple questions were directly based on construction, traversal, and structural properties.

For Dynamic Programming, focus only on standard easy-level problems such as Fibonacci, coin change (DP version), and similar 1D array-based formulations.

Graph problems are not required, as they are not mentioned in the syllabus.

Exam-Oriented Insight

The coding section tests:

- Conceptual clarity of data structures
- Clean implementation
- Understanding of time and space complexity
- Ability to write logic from scratch

The questions are straightforward if fundamentals are clear. Strong command over Trees and basic DP ensures high scoring in this section.

3 Interview Preparation

The interview preparation was structured around three major pillars:

1. Technical Depth (IT Domain)
2. Organizational Awareness (IFSCA-specific knowledge)
3. Personal and Role Justification

1. Technical Preparation (IT-Focused)

For the IT stream, the interview is heavily technical. The panel primarily evaluates:

- Depth of understanding in core IT subjects
- Practical exposure in current work profile
- Conceptual clarity in AI/ML and Cybersecurity

A significant portion of the discussion revolves around the candidate's current organization and ongoing work. Therefore, it is important to:

- Clearly explain your current role and responsibilities
- Justify architectural and technical decisions taken in your projects
- Demonstrate real-world problem-solving exposure

AI/ML-related questions are usually theoretical and application-oriented rather than mathematical. Cybersecurity questions often focus on attack vectors, mitigation strategies, and real-world implementation scenarios.

2. IFSCA-Specific Preparation

Strong institutional awareness is expected. Preparation should include:

- Overview and objectives of IFSCA
- Key provisions of the IFSCA Act, 2019
- Regulatory role in IFSC (GIFT City)
- Recent initiatives and achievements

Understanding the working of the SWIT portal (IFSCA portal) is important, including its purpose, stakeholders, and regulatory workflow.

Candidates should be able to articulate:

- How IFSCA differs from RBI, SEBI, and IRDAI
- Its unified regulatory framework within IFSC
- Its vision of making India a global financial hub

3. Personal Justification and Motivation

“Why IFSCA?” is a critical question and must be answered convincingly.

The response should reflect:

- Alignment between technical background and regulatory technology requirements
- Interest in financial technology, innovation, and policy-driven impact
- Long-term commitment rather than short-term career movement

The panel expects maturity, clarity of intent, and awareness of the responsibility associated with a regulatory role. **BE HONEST !!!**

Final Approach

The interview evaluates not only technical knowledge but also composure, clarity of thought, and institutional understanding.

A balanced preparation covering technical expertise, regulatory awareness, and personal motivation ensures strong performance in the interview.

Motivation

Have unwavering self-belief. Preparation is not only about knowledge; it is equally about confidence, resilience, and mental strength.

There will be moments of doubt, comparison, and pressure. In those moments, remind yourself that growth comes from struggle. As rightly said:

“Vo maut bhi kya jo bina lade mile, vo to phir khairat ki hui na. Jinke jeevan mein jang nahi, vo bhi bahut abhaga hote hain.”

Every meaningful achievement demands effort, discipline, and courage. The process itself transforms you.

I was able to achieve this at the age of 21. That itself proves that with clarity of focus, structured preparation, and consistency, it is absolutely possible. If I could do it, you can certainly do it as well.

Trust your preparation. Stay calm. Stay consistent. Give your absolute best.

BEST OF LUCK